THE DIVISOR OF A MEROMORPHIC SECTION PULLS BACK TO THE DIVISOR OF THE CORRESPONDING RATIONAL FUNCTION ON THE TRIVIALISING TORSOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a classification of the prime divisors of the torsor followed by a coefficient comparison through lengths over henselizations and the identification of the maximal ideals

Editorial note. The label lemma-Gm-torsor-divisor-meromorphic-section names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Divisors of meromorphic sections of an invertible sheaf and the compatibility between the two constructions of divisors, via rational functions and via lengths
- The trivialising torsor of an invertible sheaf on an integral good algebraic space, its flatness and the identification of the maximal ideals of the henselizations of the local rings of the torsor and of the base
- The length criterion for the order of a section of an invertible sheaf along a prime divisor using the total ring of fractions of the henselization, and flat base change of lengths

Source reference. Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Prerequisite pointer invoked at proof line(s) 2547; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0EPL: sources/stacks-chow/tags.txt, source lines 18381–18381. Prerequisite pointer invoked at proof line(s) 2517; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EQG: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2385–2385. Prerequisite pointer invoked at proof line(s) 2482; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof

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line(s) 2478; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0ENI: sources/stacks-chow/tags.txt, source lines 18343–18343. Prerequisite pointer invoked at proof line(s) 2495; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0ENV: sources/stacks-chow/tags.txt, source lines 18355–18355. Prerequisite pointer invoked at proof line(s) 2496; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume X is integral. Let s be a nonzero meromorphic section of $\mathcal{L}$. Let $q : T \to X$ be the morphism of Lemma 0EQG. Then*

$$q^* div_{\mathcal{L}}(s) = div_T(q^*(s))$$

where we view the pullback $q^*(s)$ as a nonzero meromorphic function on T using the isomorphism $q^*\mathcal{L} \to \mathcal{O}_T$

Proof. Observe that $\mathrm{div}_T(q^*(s)) = \mathrm{div}_{\mathcal{O}_T}(q^*(s))$ by the compatibility between the constructions given in Spaces over Fields, Sections 0ENI and 0ENV. We will show the agreement with $\mathrm{div}_{\mathcal{O}_T}(q^*(s))$ in this proof. We will use all the properties of $q : T \to X$ stated in Lemma 0EQG without further mention. Let $Z \subset T$ be a prime divisor. Then either $Z \to X$ is dominant or $Z = q^{-1}(Z')$ for some prime divisor $Z' \subset X$. If $Z \to X$ is dominant, then the coefficient of Z in either side of the equality of the lemma is zero. Thus we may assume $Z = q^{-1}(Z')$ where $Z' \subset X$ is a prime divisor. Let $\xi' \in |Z'|$ and $\xi \in |Z|$ be the generic points. Then we obtain a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(\mathcal{O}^h_{T,\xi}) & \xrightarrow{c_\xi} & T \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle q} \\ \mathrm{Spec}(\mathcal{O}^h_{X,\xi'}) & \xrightarrow{c_{\xi'}} & X \end{array}$$

see Decent Spaces, Remark 0EPL. Choose a trivialization $s_{\xi'}$ of $\mathcal{L}_{\xi'} = c^*_{\xi'}\mathcal{L}$. Then we can use the pullback s_ξ of $s_{\xi'}$ via h as our trivialization of $\mathcal{L}_\xi = c^*_\xi q^*\mathcal{L}$. Write $s/s_{\xi'} = a/b$ for $a, b \in \mathcal{O}_{X,\xi'}$ nonzerodivisors. By definition the coefficient of Z' in $\mathrm{div}_{\mathcal{L}}(s)$ is

$$\mathrm{length}_{\mathcal{O}^h_{X,\xi'}}(\mathcal{O}^h_{X,\xi'}/a\mathcal{O}^h_{X,\xi'}) - \mathrm{length}_{\mathcal{O}^h_{X,\xi'}}(\mathcal{O}^h_{X,\xi'}/b\mathcal{O}^h_{X,\xi'})$$

Since $Z = q^{-1}(Z')$, this is also the coefficient of Z in $q^*\mathrm{div}_{\mathcal{L}}(s)$. Since $\mathcal{O}^h_{X,\xi'} \to \mathcal{O}^h_{T,\xi}$ is flat the elements a, b map to nonzerodivisors in $\mathcal{O}^h_{T,\xi}$. Thus $q^*(s)/s_\xi = a/b$ in the total quotient ring of $\mathcal{O}^h_{T,\xi}$. By definition the coefficient of Z in $\mathrm{div}_T(q^*(s))$ is

$$\mathrm{length}_{\mathcal{O}^h_{T,\xi}}(\mathcal{O}^h_{T,\xi}/a\mathcal{O}^h_{T,\xi}) - \mathrm{length}_{\mathcal{O}^h_{T,\xi}}(\mathcal{O}^h_{T,\xi}/b\mathcal{O}^h_{T,\xi})$$

The proof is finished because these lengths are the same as before by Algebra, Lemma 02M1 and the fact that $\mathfrak{m}^h_\xi = \mathfrak{m}^h_{\xi'}\mathcal{O}^h_{T,\xi}$ shown in Lemma 0EQG. □